

Jayashan Manodya

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ABOUT ME

I am an IT undergraduate specializing in Artificial Intelligence with a strong passion for becoming an AI Engineer and building intelligent technologies that solve real-world problems. I am currently focused on Generative AI, machine learning, computer vision, RAG systems, and full-stack development while continuously improving my skills through hands-on projects and modern software engineering practices. I have also developed agentic AI projects using multi-agent architectures and orchestration frameworks for intelligent workflow automation and context-aware decision-making. I enjoy developing scalable AI-powered systems and continuously exploring emerging technologies in AI and software engineering.

EDUCATION

- **BSc (Hons) - IT Specializing in Artificial intelligence (In Progress)** (2024 - Present)
Sri Lanka Institute of Information Technology
Most Recent Semester[Y2S2] GPA: 3.93/4.0
- **G.C.E Advanced Level** (2020 - 2023)
Prince College - Ratnapura
Physical Science with ICT (Z-Score : +0.84) | Index : 7371365

PROJECTS

- **Artificial Intelligence**
 - **KIKO - Kapuruka Shopping Agent** [PREVIEW](#) [DEMO](#)
KIKO is a AI-powered conversational shopping assistant with a **5-agent architecture** using the **Kapruka MCP**. Enabled users to **browse categories, search products, view product details, validate delivery, manage carts, complete guest checkout, and track orders** entirely through natural **language conversations in English, Sinhala, and Tamil**. Implemented intelligent agent routing, tool orchestration, self-verification, and dynamic UI rendering to create a scalable and trustworthy end-to-end shopping platform.
[Python](#) [FastAPI](#) [LangChain](#) [LangGraph](#) [Next.js](#) [Open AI](#) [Kapruka MCP](#) [PostgreSQL](#)
 - **IKMS - RAG-Powered Custom Document Question Answering System** [DEMO](#)
Developed a Retrieval-Augmented Generation (RAG) document question-answering system that enables users to upload PDFs and receive accurate, context-aware answers. Built with FastAPI, LangGraph, OpenAI GPT-4o, and Pinecone, the solution uses a multi-agent pipeline for query planning, document retrieval, answer generation, and verification, ensuring reliable and grounded responses with scalable serverless deployment.
[Python](#) [FastAPI](#) [LangChain](#) [LangGraph](#) [Pinecone](#) [PyPDF](#) [React](#)
 - **PlateX - AI License Plate Detection & Recognition System**
Developed an AI-powered license plate recognition system capable of real-time vehicle plate detection and automated text extraction from images. Implemented OCR, computer vision preprocessing, and data export functionalities to improve detection accuracy, automate vehicle identification workflows, and provide an interactive web-based processing interface.
[Python](#) [YOLOv12](#) [PaddleOCR](#) [Flask](#) [OpenCV](#)
 - **WeatherLK - ML Weather Prediction System**
Created an ML-powered weather forecasting platform capable of predicting temperature, rainfall, and wind conditions using historical and environmental data. Implemented machine learning models, data preprocessing pipelines, and real-time forecast visualization features to deliver accurate weather insights through an interactive web application.
[Python](#) [FastAPI](#) [RandomForestRegressor](#) [FastAPI](#) [NumPy](#) [React](#)
- **Full Stack**
 - **TeaPlanter (Smart Tea Estate Management Platform)** [DEMO](#)
Developed an AI-powered tea estate management platform for handling workforce, harvest, attendance, inventory, payroll, delivery, and financial operations within plantation environments. The platform includes role-based dashboards for owners, clerks, and workers with real-time operational monitoring and management features. Integrated AI-powered tea leaf disease detection, machine learning-based tea price prediction, and an intelligent multi-agent AI assistant to support agricultural monitoring, automation, productivity analysis, and data-driven decision-making for modern tea plantation management.
[Java](#) [Spring Boot](#) [PostgreSQL](#) [React](#) [TypeScript](#) [Tailwind CSS](#) [Spring Security](#) [OAuth2](#) [Clerk](#) [YoloV8](#) [Random Forest](#)
 - **Fuel Alert - Fuel Availability & Queue Tracking.** [DEMO](#)
Built a real-time fuel availability and queue tracking platform with live station monitoring, queue estimation, geofencing validation, and multilingual accessibility features. Developed secure administrative management functionalities and optimized the system for mobile access to improve user convenience and real-time fuel distribution awareness.
[React](#) [JavaScript](#) [TypeScript](#) [Node.js](#) [Express.js](#) [PostgreSQL](#) [Leaflet](#)

RailLink - Train Scheduling & Booking System - Module Project (Team)

AI-assisted railway management and booking platform featuring secure ticket reservations, automated notifications, administrative controls, and an intelligent chatbot to optimize transportation workflows and passenger support.

Java Spring Boot MYSQL React JavaScript TypeScript Thymeleaf Spring Security n8n Maven

Mobile Development

BookLink - Community Book Exchange Mobile Application

community-driven mobile platform for book exchanges, featuring user authentication, real-time chat, wishlist management, review systems

React Native Expo Express.js MongoDB Clerk Node.js JavaScript TypeScript

Electronic

Automated Room Comfort System

Developed an IoT-based room comfort system using microcontrollers, adjusting temperature and lighting based on sensor data

Arduino C++ Micro Controllers ESP8266 DHT11 BH1750

Featured selected projects. More available on [GitHub](#).

EXPERIENCE

Work Experience

Creative Designer(Part Time) - Glanz Digital (PVT) Ltd

(2025 - Present)

Currently working part-time as a Creative Designer at Glanz Digital, designing social media posts, promotional flyers, branding materials, digital advertisements, and marketing visuals while collaborating with creative and marketing teams on client-focused digital campaigns and brand identity development.

Extra-Curricular Activities

IEEE Computer Society - SLIIT Student Branch

Member

(2024 - Present)

Senior Prefect - Prince College

Member

(2018 - 2020)

Aviation Society - Prince College

(2019 - 2020)

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C, C++
- **Frameworks & Libraries:** FastAPI, Spring Boot, Node.js, React, LangChain, LangGraph, Flask, Express.js, React Native,
- **AI & ML Tools:** AI, Machine Learning, Generative AI, RAG Systems, AI Agents, LLM-based Applications, OpenAI APIs, Scikit-learn, YOLO, PaddleOCR, OpenCV, Pandas, NumPy
- **Databases:** PostgreSQL, MongoDB, Vector Databases, MySQL, File-based Storage Systems
- **Dev Tools :** Git, GitHub, Maven, n8n
- **Deployment & Cloud :** Railway, Vercel, Render

CERTIFICATIONS

- AI Engineer Bootcamp – STEM Link
- AI/ML Engineer – Stage 1 – SLIIT
- Programming in Python – CODL, University of Moratuwa
- Web Development – CODL, University of Moratuwa
- Fundamentals of Digital Marketing – Google Digital Garage
- Arduino CodeCamp Workshop – CODECAMP

ACHIEVEMENTS

- Dean's List Recipient - [Y2S2] : Achieved semester GPA above 3.7
- Ranked 11th Place – Bashaway 2024 (Team: Nyx).
- Top 20 Finalist & Exhibitionist – IoT Project Showcase Awards/Activities

REFERENCES

Mr. Amila Nuwan Alexander

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